

Project Ideas

Shortlist focused on realistic projects with good chances of strong results

Quick Ranking

#	Idea	Dataset	Effort	Why it stands out
1	Plant disease classification + Grad-CAM	PlantVillage	Low	Best balance between strong metrics, simple setup, and enough substance for the final report.
2	Steel surface defect classification	NEU-CLS	Very low	Probably the safest option if we want something compact, quick to iterate on, and still meaningful.
3	Traffic sign classification	GTSRB	Low	Very clean benchmark with good labels, strong baselines, and room for robustness experiments.
4	Satellite land-use classification	EuroSAT	Low	Easy to train, visually interesting, and a bit more distinctive than a generic image dataset.
5	Pneumonia classification from chest X-rays	Kermany Chest X-Ray	Low	Good demo value and clear metrics, but the ethics and limitations discussion needs more care.

1. Plant Disease Classification + Grad-CAM

It is a very safe project technically, but it still gives enough room to discuss model choice, explainability, class imbalance, and dataset bias.

- **Task:** classify plant leaf images into disease categories.
- **Dataset:** PlantVillage, around 54k images and 38 classes.
- **Model idea:** start with a simple CNN baseline, then fine-tune something light like ResNet18, MobileNetV3, or EfficientNet-B0.
- **Evaluation:** accuracy, macro-F1, confusion matrix, per-class recall, and a few Grad-CAM visualizations.
- **Why it is manageable:** public dataset, straightforward preprocessing, and transfer learning should already give strong results.
- **Main risk:** the dataset is relatively clean and controlled, so real-world generalization may be weaker than the test results suggest.

2. Steel Surface Defect Classification

It is small, practical, and probably the fastest project to get working well.

- **Task:** classify steel surface images into defect categories.
- **Dataset:** NEU-CLS, 1,800 grayscale images across 6 classes.

- **Model idea:** simple CNN baseline, then ResNet18 or MobileNetV3 after converting grayscale to 3 channels.
- **Evaluation:** accuracy, macro-F1, per-class recall, and ideally cross-validation because the dataset is small.
- **Why it is manageable:** fast training, small dataset, clear labels, and an industrial application that still looks serious.
- **Main risk:** because the dataset is small and visually consistent, the model may overfit to that specific setting.

3. Traffic Sign Classification

It is a classic benchmark, which makes the experimental setup very easy to justify.

- **Task:** classify cropped traffic sign images.
- **Dataset:** GTSRB, more than 50k images and 40+ classes.
- **Model idea:** custom CNN as baseline, then ResNet18 or EfficientNet-B0.
- **Evaluation:** accuracy, macro-F1, confusion matrix, and maybe one robustness experiment with blur or brightness changes.
- **Why it is manageable:** great dataset quality, lots of benchmark history, and no complicated pipeline.
- **Main risk:** it is a bit standard, so the project becomes more interesting if we add robustness or efficiency analysis.

4. Satellite Land-Use Classification

It still feels simple, but the application domain is a bit more distinctive.

- **Task:** classify land-use categories from satellite image patches.
- **Dataset:** EuroSAT, 27k labeled samples in 10 classes.
- **Model idea:** ResNet18 or MobileNetV3, with EfficientNet-B0 as an optional extra comparison.
- **Evaluation:** accuracy, macro-F1, confusion matrix, and error analysis for visually similar classes.
- **Why it is manageable:** public dataset, low compute requirements, and easy transfer learning setup.
- **Main risk:** it can become too straightforward, so the report should include a bit of analysis beyond raw accuracy.

5. Pneumonia Classification from Chest X-Rays

It has strong presentation value and very clear evaluation metrics, even with a simple binary classifier.

- **Task:** classify chest X-rays as normal or pneumonia.

- **Dataset:** Kermany Chest X-Ray subset, around 5.8k labeled images.
- **Model idea:** simple CNN baseline, then ResNet18 or DenseNet121, plus optional Grad-CAM.
- **Evaluation:** accuracy, precision, recall, F1, AUROC, and special attention to false negatives.
- **Why it is manageable:** binary classification keeps the pipeline simple and the metrics easy to interpret.
- **Main risk:** this is the one that requires the most careful wording around bias, shortcuts, and real-world medical use.

Best Picks Depending on What We Want

- **Safest overall choice:** PlantVillage.
- **Smallest implementation burden:** NEU-CLS.
- **Best balance between simple and a bit more distinctive:** EuroSAT.
- **Best benchmark-style project:** GTSRB.
- **Best for a strong demo narrative:** Chest X-rays.

Useful Links

- PlantVillage: https://www.tensorflow.org/datasets/catalog/plant_village
- NEU-CLS: <https://doi.org/10.6084/m9.figshare.28903550>
- GTSRB: https://benchmark.ini.rub.de/gtsrb_dataset.html
- EuroSAT: <https://github.com/phelber/EuroSAT>
- Chest X-Ray dataset: <https://data.mendeley.com/datasets/rscbjbr9sj/3>